POLICIES FOR SENIOR SEMINAR

1. Objectives of course

To give students the opportunity to do in-depth and independent mathematics-related research projects that would require students to integrate their mathematical knowledge from one or more different areas.

To give students the opportunity to enhance their communication skills by way of written reports and oral presentations.

2. Overview

Welcome to the math discipline senior seminar! In this seminar, you will be working on a single topic throughout the academic year, writing a paper on it, and presenting your topic to an audience consisting of at least the Math Faculty and your peers in this course.

You will be meeting regularly with your advisor(s), member(s) of the math faculty, throughout the process.

In addition to your advisor, it is strongly recommended that you have a second reader for your paper so that you receive constructive suggestions from more than one person (advisor) way before you submit your near-final version to the mathematics faculty. This second reader should be another (preferably math) faculty member.

You should give a version of your paper to your second reader at least two weeks before your near final version is due. The idea is that your near final version has been read and commented on by at least two faculties before the other math faculty members read them.

We suggest that you have your second reader chosen at the beginning of the second semester of your project so that he/she is aware of the time required to read your paper.

3. Your Senior Seminar (in general)

The mathematics senior seminar is the capstone course in the mathematics curriculum, and it is meant to be an integrative experience for you. You may be combining topics, materials, and skills you learned from more than one class, or you may be combining your mathematical knowledge with a personal interest. Certainly the paper will require you to use your writing skills as well as your mathematical skill. For a handful of students, the presentation is a totally new experience, requiring mastery of the subject matter, technical control over the mode of presentation, and public speaking skills. We are sure you will find your year-long experience challenging while at the same time, fun and worthwhile.

Your topic

You will be spending a lot of time on studying your topic, so make sure you have a topic that interests you! Topics in pure mathematics or applied mathematics are all very welcome. A topic directly related to something in a course you have taken or are taking would be a good idea.

Your paper

Papers are typically about 10-15 single-spaced pages in length, including small figures and tables, if any. Some papers may have lengthy appendices containing data, computer-generated tables, etc.

You must include your name, your faculty advisor's name, your second reader's name, the title of project, date, and an ABSTRACT, at the beginning of your paper. (More details about your paper are given below.)

Template for math styles and format for writing your paper

Your presentation

Presentations are 15 minutes in length, the first 5 minutes of which should be geared towards a general audience; a good suggestion for the 1st 5 minutes would be to touch on the motivation or the general mathematical ideas behind your project. The audience will include the mathematics faculty members and the other students in the seminar.

4. Details and timeframe

Similar to last academic year, there will be a time-line that a student can follow to complete MAT 492. The total length of time (from the beginning till the final report is turned in) for a student to work on senior paper is "two semesters".

Only under extenuating circumstances will other arrangements be made. The coordinator of MAT 492 needs to be informed of any changes or extenuating circumstances at the beginning of the semester you plan to finish your senior seminar, i.e. your presentation.

If you do not follow the deadlines, you risk receiving an F grade for MAT 492.

MORE INFORMATION ABOUT THE PROPOSAL, NEAR FINAL VERSION, DISCUSSION WITH FACULTY, AND FINAL VERSION IS GIVEN BELOW.

5. Your proposal

During the first semester that you signed up for MAT 492, you are expected to do preliminary work, meaning searching for the math literature and looking for the mathematical foundation needed for your project. Your proposal should be a summary of your project's basic intent, what you have done, and

what you have yet to do to complete your senior seminar project. You MUST submit an electronic copy of your proposal to the coordinator by the due date.

6. The Near-Final version of your Paper

By definition, near-final means this version is almost an ε amount away from the final version. Your near final version should include suggestions from your advisor and your second reader; and it should contain the name of your advisor and second reader, and an abstract.

Your near final version should not have sections or subsections undone nor the entire set of references/bibliographies missing.

The difference between the near-final version and the final version includes suggestions from your advisor, the feedback from faculty during the 15-min discussion, and the feedback from anyone during the presentation.

You must submit an electronic copy of your near final version to the coordinator by the due date.

7. Discussion (15 minutes) with faculty committee

At this discussion, the faculty committee will give you their responses to the near final version of your reports and they will give you a few suggestions or even have questions for you. Usually, it takes about 15 minutes per student for this discussion.

You are requested to incorporate changes or suggestions from the faculty committee into your final version of your report and also into your presentation.

8. The Final version of your Paper

You are advised to incorporate as many appropriate suggestions as possible from your advisor, the feedback from faculty during the 15-min discussion, and the feedback from anyone during the presentation into your final version.

You must include your name, your faculty advisor's name, your second reader's name, the title of project, course (MAT 492 Math Senior Seminar), date, and an abstract, at the beginning of your paper.

You must submit an electronic copy of your final version to the coordinator by the due date.

9. Advisory comments on procedure

All students need to register for MAT 492 for the semester that they want to start their senior seminar. You will be receiving a “A” grade at the end of the semester, unless you have shown abysmal progress, or did not turn in your math proposal on time, or just plain slacked off.

Since your choice of advisor and general area is important to you, you should probably actually make these choices during Christmas break or summer break if you are starting in Spring or Fall semesters, respectively.

You are expected to meet about twice a month with your advisor throughout the academic year. Towards the end of the first semester, your advisor will expect to be seeing appropriate progress toward completion of the paper. You should be becoming clearly more expert on your topic. You should be gathering the necessary data or carrying out the necessary computer work, if appropriate. The paper should be taking shape. By the middle of the second semester, you should start preparing your presentation as well.

If by the end of your first semester in MAT 491, you did not show any progress in your senior paper, or have not identified a math faculty as an advisor, or have not been working with your advisor, or did not turn in your math proposal, then you will earn a grade of F at the end of the first semester.

Unless prior arrangements have been made with the MAT 491/492 coordinator, if by the end of your second semester in MAT 492, you did not complete your presentation, or did not turn in your final paper on time, then you will earn a grade of F.

10. Your grade

GRADING SCHEME

30% Active participation throughout the process

10% Project Proposal with mathematical foundation and research plans

30% Final written paper

30% 15-min presentation

Active participation throughout the process means primarily that you have worked substantially in between meetings with your advisor and you have attended all the math senior seminar presentations during the two semesters that you are working on your senior seminar project.

Your overall course grade will be decided by your advisor in consultation with the rest of the math discipline faculty.

Your presentation will be assessed by the math faculty who will use the following rubrics and criteria as a guide.

To enhance the value of the presentation component of the course, audience members will fill out the MAT 492 evaluation forms after each presentation. Your advisor may use these comment forms to help determine your grade for your presentation. However, the principal purpose of these forms is to provide you with useful feedback.

11. Specific guidelines for paper and presentation

Periodic feedback on your progress:
As mentioned in the overview section, you will be meeting regularly with your advisor(s), member(s) of the math faculty, throughout the process of doing your project and writing up your paper.

In addition to your advisor, you should have a second reader for your paper so that you receive constructive suggestions from more than one person (advisor) way before you submit your near-final version to the mathematics faculty. This second reader should be another (preferably math) faculty member. And second reader should have a draft of your paper at least two weeks before the near final version is due.

Originality:
You should strive for some degree of originality in your project.

In pure mathematics, if you choose to present a famous result, you might find it hard to be original. In this case you might want to examine the presentation of this result in two reference works. You could combine what you felt was best from these two presentations, keeping in mind your particular audience. You could perhaps add an example that you worked out yourself.

In applied mathematics, applying something you learned in a class to your own situation or your own data would be original enough.

Level:
In your paper and especially your presentation, you should aim to be understandable to your fellow seminar members. It is easy to have thought about the basics of your topic for so long that you forget that others are just beginning in your topic. "Blowing away'' the audience by formula after formula in your presentation is severely frowned upon!

Mathematical content:
Many topics, especially in applied mathematics, have both a substantial non-mathematical component and a substantial mathematical component. This being the math discipline seminar, you should make sure that the mathematical component of your topic gets considerable attention.

Regardless of your topics and of the areas of your topics, if you are actually using what you learned in some of your advanced math courses here, you're probably on track. What to include and what to omit in your paper and presentation is subtle; your advisor should give you help here.

Professional quality
Your final paper and your presentation should be quite polished. You are expected to use LaTeX, Word, or some other high quality software for the paper. The visual component of your presentation can be a mixture of blackboard work, transparencies, and computer projection. In most cases, the best choice is to work primarily or exclusively with transparencies. It is best if the spoken part of your presentation has a sense of informality to it. Memorizing and repeating your written paper is not appropriate! On the

contrary, you can expect and should welcome questions that might take you a bit to one side of what you planned to say. Again your advisor should be able to help you out on all these issues.

Attributes of a Math Paper

By and large, math papers or technical papers follow some systematic organizations, nomenclatures, and styles.

- You should include an abstract in your paper.
- You are to always define a term before using it in the text.
- Figures should be properly labeled and correctly referenced from the text.
- Bibliographies should be included as a section and should be referenced within the text.
- The paper should be separated or organized into sections, each of which with appropriate subtitles.